Grouping in pairs

1

a $(5+v)(a+b)$

b $(y-z)(x-w)$

c $(4w+3x)(5y-z)$

d $\left(2x^2+3z\right)(2y-1)$

e $(g+h)(3f+g+h)$

f $(y-4)(8y-3)$

g $(2c-d)(c+5d+3)$

h $\left(3x^2-5y\right)(x+4y)$

2

a $(x-2y)(8+z)$

b $(3+x)(8+y)$

c $(x+z)(7y+w)$

d $(x+2)(x+5)$

e $(x-3)(x+8)$

f $(p+2)(2m+3)$

g $(a+8)\left(a^2+1\right)$

h $(pq+rs)(3q-11y)$

i $(4x+3z)(1+8y)$

j $(2x+3z)(1+6y)$

Perfect squares

3

a $(q+t)^2$

b $(u-q)^2$

c $(b-r)^2$

d $(x+3)^2$

e $(c-2)^2$

f $(8+e)^2$

g $(7-p)^2$

h $(4w-5)^2$

i $(8r+3)^2$

j $\left(\frac{1}{2}-3d\right)^2$

4

a $\left(x^2+4\right)^2$

b $(a-3)^2(a+3)^2$

5

a $t+7$

b $4(t+7)$

Difference of two squares

6 $x+4$

7 $z-7$

8

a $(x+y)(x-y)$

b $(n-5)(n+5)$

c $(v-1)(v+1)$

d $(11-v)(11+v)$

e $\left(x-\frac{1}{2}\right)\left(x+\frac{1}{2}\right)$

f $\left(x-\frac{5}{11}\right)\left(x+\frac{5}{11}\right)$

g $(4+3y)(4-3y)$

h $x^2y^2-49=(xy-7)(xy+7)$

i $(5m+7)(5m-7)$

j $(9x+4y)(9x-4y)$

k $7x^2-63=7(x-3)(x+3)$

l $5(3t+2)(3t-2)$

9

a 256 b 392 c 261 d 660

10

a $24\,696$ b $89\,984$

11

a x^2-6^2

b

i No ii No iii Yes

Mixed factorisations

12

a $3y\left(x^2-2y^3+3x^2y^2\right)$

b $(4xy+3z)^2$

c $(11c-7d)(11c+7d)$

d $\left(x^2+y\right)(3y+z)$

e $(a+b-c)(a+b+c)$

f $4b\left(2a^2b^3+3c^2\right)$

g $7(2b+1)$

h $7hp^2\left(2h^2r+3h^4+7p^2\right)$

i $(8w+3t)\left(w-4t^2\right)$

j $\left(4q^2+3r\right)^2$

k $3d(4f-h)(4f+h)$

l $2t\left(8t-3s^2\right)\left(8t+3s^2\right)$

m $2(3j+5k)^2$

n $(x-1)(x+1)(y+7)$

13 $(x-1)(x+1)\left(x^2+1\right)$